PlateView

An AR Interactive Menu, Ordering & Billing Experience

Product Design Document (PDA)
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1. Executive Summary

PlateView is a restaurant menu application that replaces flat food photographs with augmented reality (AR) previews. Instead of guessing what a dish looks like from a name and a static photo, a diner taps a menu item, points their phone camera at the table, and sees the dish appear life-size right in front of them — true-to-scale, walk-around-able, plating and portion included. Where AR isn't available or practical, the same 3D model can still be viewed as a rotatable on-screen object. The app then lets the diner add the item to an order, customize it, and pay the bill, all from the table.

The core bet is simple: people order better, order more confidently, and are more satisfied with what arrives when they can actually see what they're getting — at real scale, on their own table — before they commit. The AR view is the differentiator; ordering and billing are the features that turn that differentiator into a complete, sellable product for restaurants.

2. Problem Statement

Traditional menus — printed or digital — describe food with text and, at best, a single static photo shot under studio lighting that rarely matches what's served. This creates friction for diners and lost revenue for restaurants:

- **Diners order blind.** Descriptions like “pan-seared, herb-crusted, seasonal reduction” don't convey portion size, spice level, or plating — leading to hesitation, over-ordering, or disappointment.
- **Photos mislead.** A single angle under ideal lighting sets expectations that the actual dish can't meet, driving complaints and lower ratings.
- **Menu decisions take too long.** Large parties spend significant time deliberating, asking servers to describe dishes, which slows table turnover.
- **Ordering and payment are still manual** in most sit-down restaurants, adding wait time on both ends of the meal.

3. Vision & Goals

Vision: Make ordering food as visual and confident as seeing it on the table in front of you — point, look, understand it, then order it.

Primary goals for v1:

- Let diners view any menu item in AR, placed life-size on their own table, before ordering.
- Let diners build an order directly from the AR/3D menu view.
- Let diners pay the bill in-app, itemized and split-friendly.
- Give restaurants a simple way to add/manage menu items and 3D assets without engineering help.

4. Target Users

Persona	Who they are	What they need
Dine-in Diner	Guest seated at a restaurant table, browsing the menu on their own phone via QR code or a table tablet.	Fast, visual way to decide what to order without flagging down a server; confidence that the dish matches expectations.
First-time / Tourist Diner	Unfamiliar with the cuisine or specific dish names on the menu.	Visual context for unfamiliar dishes (e.g. regional or ethnic cuisine) to reduce ordering anxiety.
Restaurant Owner / Manager	Runs a single restaurant or small chain; owns the menu content.	Simple tools to upload/manage menu items, prices, and 3D assets; visibility into orders and sales.
Server / Floor Staff	Supports tables, handles exceptions the app can't.	A system that reduces repetitive “what's in this dish” questions and speeds up order-to-kitchen flow.
Kitchen / Back of House	Prepares food based on incoming orders.	Clear, structured order tickets (item, customization, quantity) fed from the app without re-keying.

5. Core Concept & User Journey

The experience is built around one interaction loop: **browse → preview in AR → customize → add to order → confirm → pay.**

Step	What happens
1. Access the menu	Diner scans a table QR code (or opens the app if the restaurant is saved/favorited). No login required to browse.
2. Browse categories	Menu is organized by category (starters, mains, desserts, drinks) with short text + price, similar to today's digital menus.
3. Tap an item	Item opens the camera and drops the dish onto the table at real-world scale (AR mode). The diner can walk around it, zoom in, and view it from any angle. On devices/browsers without AR support, the same model opens in an on-screen rotatable 3D viewer instead.
4. Read details	Below/beside the AR/3D view: ingredients, allergens, spice level, calories (if provided), and customization options (e.g. “no onions,” “extra spicy”).
5. Add to order	Diner sets quantity and customizations, adds to a running cart shared across the table.
6. Review shared cart	Everyone at the table sees the combined order in real time before it's sent to the kitchen.
7. Send to kitchen	Order is submitted; kitchen receives a structured ticket. Diner sees order status (received / preparing / served).
8. Bill & pay	At any point, the diner can view the itemized bill, split it (by item or evenly), tip, and pay in-app.

6. Key Features

6.1 Digital Menu

- Category-based browsing with search and dietary filters (vegetarian, vegan, gluten-free, allergens, spice level).
- Text description, price, and prep-time estimate shown for every item.
- Works via QR code at the table — no app install required for the core browsing/viewing flow (app install unlocks saved favorites, order history, faster repeat ordering).

6.2 AR Food Preview (core differentiator)

- Tapping a menu item launches the phone's camera and places a true-to-scale 3D model of the dish directly on the diner's table — real portion size, real plating, viewable from any angle by moving the phone or walking around it.
- Built on standard mobile AR frameworks (ARKit on iOS, ARCore on Android) surfaced through a web-based AR viewer, so it works straight from the QR-code menu without an app install.

- Automatic fallback to an on-screen rotatable 3D viewer (orbit, pinch-zoom, auto-rotate) on devices or browsers that don't support AR, or when the diner prefers not to use the camera — the experience is AR-first, not AR-only.
- Progressive loading (low-res placeholder → full model) so the AR view feels near-instant even on slower restaurant Wi-Fi.
- Fallback: if a 3D/AR model isn't available yet for an item, the app shows a high-quality photo instead — the system should never block ordering on missing 3D content.

6.3 Ordering

- Add to cart directly from the AR/3D view, with customization options (e.g. spice level, remove ingredient, add-ons) surfaced as simple toggles/dropdowns.
- Shared table cart: multiple diners at the same table can add items to one order from their own phones.
- Order sent straight to a kitchen display / printer as a structured ticket — no manual re-entry by staff.
- Live order status for the diner (received, in the kitchen, served).

6.4 Billing & Payment

- Itemized, real-time bill that updates as items are added.
- Split the bill evenly or by item/person.
- In-app payment (card / wallet) with tipping built in.
- Digital receipt emailed or shown in-app; optional order history for repeat customers.

6.5 Restaurant Admin Dashboard

- Web dashboard for restaurant staff to add/edit menu items, prices, categories, and availability (86'd items marked out of stock instantly).
- Upload flow for 3D assets (see Section 7) and fallback photos.
- Basic sales and popularity analytics (best-sellers, peak order times, average ticket size).

7. Technical Approach

7.1 Getting 3D Models of Food (the hard part)

This is the biggest open technical question and the thing worth validating first. Three realistic paths, roughly in order of practicality for a v1:

Approach	How it works	Trade-offs
Photogrammetry capture	Restaurant (or a PlateView field team) photographs each dish from multiple angles with a phone rig; software (e.g. RealityCapture, Polycam, Luma AI) reconstructs a 3D mesh.	Most realistic results; requires a one-time capture session per dish and re-capture if plating changes. Best fit for a pilot with a small number of partner restaurants.
Stock / library 3D assets	License or commission a library of generic 3D food models (e.g. “margherita pizza,” “caesar salad”) and map menu items to the closest match.	Fast to launch, low cost, scales across restaurants instantly — but less accurate to the specific dish, which partly undercuts the value proposition. Good stopgap for less differentiated items.
AI-generated 3D from photos	Use image-to-3D generation models to turn a restaurant's existing food photography into an approximate 3D asset automatically.	Cheapest to scale, improving quickly as a technology, but quality/accuracy still varies by dish type (liquids, sauces, and irregular shapes are harder). Worth prototyping but not betting the MVP on.

Recommendation: start with photogrammetry capture for a small pilot menu (10–20 signature dishes at 1–2 partner restaurants) to prove the experience is actually more useful than a good photo. Use stock assets or photos as fallback for everything else so no item is ever blocked from being orderable.

7.2 Platform & Stack (suggested)

Layer	Suggested approach
Diner client	Mobile-first web app (PWA) accessed via QR code, so there's no install friction at the table; native iOS/Android app later for saved favorites, order history, and push notifications.
AR / 3D rendering	Web-based AR viewer (e.g. <model-viewer> with AR Quick Look on iOS and Scene Viewer on Android, or WebXR) using compressed .glb/.gltf/.usdz model files for fast mobile loading; same asset also drives the on-screen 3D fallback viewer.
Ordering & kitchen sync	Real-time backend (websockets) pushing structured order tickets to a kitchen display system or receipt printer.
Payments	Integrate an established payments provider (e.g. Stripe) for card/wallet payments, tipping, and split billing — avoid building payment processing in-house.

Layer	Suggested approach
Admin dashboard	Web app for restaurant staff to manage menu, pricing, availability, and 3D/photo assets.
Asset pipeline	Backend service to ingest, compress, and version 3D model files per menu item.

7.3 Non-Functional Requirements

- **Performance:** 3D model must start rendering in under ~2 seconds on average restaurant Wi-Fi; use progressive loading and aggressive compression.
- **Reliability:** Ordering and payment flows must degrade gracefully — if 3D assets fail to load, the diner can still see a photo, description, and order normally.
- **Accessibility:** Full menu content (name, description, price, allergens) must be available in plain text/screen-reader-friendly form, not locked behind the AR/3D viewer.
- **Security:** Payment data handled entirely by the payments provider (PCI compliance offloaded, not handled in-house).

8. Success Metrics

Metric	Why it matters
AR view engagement rate	% of menu views where the diner launches AR (vs. skipping straight to text) — validates that the feature is actually used, not just present.
Conversion from view to order	% of viewed items that get added to the order — core signal that seeing the dish increases ordering confidence.
Average order value	Compare tables using PlateView vs. traditional menu to test the “see it, want it more” hypothesis.
Order accuracy / complaint rate	Fewer “this isn't what I expected” complaints and kitchen remakes is a direct measure of the core value proposition.
Table turnover time	Faster decision-making and in-app payment should reduce total table occupancy time.
Restaurant retention	% of pilot restaurants that continue past the trial period — proves the business, not just the diner experience, works.

9. Phased Roadmap

Phase	Scope
Phase 0 — Validate	Build a clickable AR prototype with 10–20 3D-modeled dishes (no live ordering) and test with real diners at 1 partner restaurant. Goal: confirm the AR preview meaningfully changes ordering behavior before investing in the full build.
Phase 1 — MVP	QR-code digital menu with AR preview (plus on-screen 3D fallback) for a pilot menu, basic cart/ordering to kitchen ticket, and in-app payment. Single restaurant or small chain pilot.
Phase 2 — Core product	Restaurant admin dashboard for self-serve menu/asset management, shared table cart, order status tracking, split billing, basic analytics.
Phase 3 — Scale	Onboarding flow for new restaurants (including a repeatable 3D-capture or AI-generation pipeline), native mobile app with saved AR favorites, loyalty, multi-location support.
Phase 4 — Expand	Multi-item AR table layout (preview a whole order placed together), personalized recommendations, integration with existing POS systems used by restaurants.

10. Risks & Open Questions

Risk / Question	Notes
Cost and effort of producing 3D models at scale	This is the make-or-break constraint. Needs a cheap, repeatable capture or generation pipeline before this can scale beyond a handful of pilot restaurants.
Does an AR preview actually beat a good photo?	Should be tested directly in Phase 0 before building ordering/billing — AR is only worth the cost if it measurably changes ordering behavior.
AR hardware/browser support varies	Not every phone or browser supports WebXR/AR Quick Look/Scene Viewer well. The on-screen 3D fallback must feel like a real experience, not a degraded one, for the sizeable share of diners who'll land there.
Restaurant willingness to adopt new tech at the table	Staff training, device provisioning (QR vs. tablets), and integration with existing POS/kitchen systems all affect adoption.
Menu changes and freshness	Restaurants change menus/specials often; the admin tooling and asset pipeline must make updates fast, or content will go stale.
Payment compliance	Offload PCI compliance to a payments provider rather than handling card data directly.
Connectivity at the venue	Many restaurants have weak Wi-Fi; the app must work acceptably on cellular data and degrade gracefully when 3D assets are slow to load.

This document is a first draft intended to align on scope and sequence a validation plan — not a final spec. Next step: build the Phase 0 prototype and test the core AR-preview hypothesis before committing engineering resources to ordering and billing.